COMPUTER NETWORKS

Designing a Redundant Network Topology

Implementing HSRP (Hot Standby Router Protocol)

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Importance of Redundancy

Business Continuity

In modern enterprise networks, even seconds of downtime can lead to significant financial and operational losses.

Redundancy ensures high availability by eliminating single points of failure (SPOF) at the gateway level.

- 99.999% Network Uptime
- Seamless Link Failover
- Fault Tolerant Architecture

Zero-Downtime Design

What is HSRP?

Hot Standby Router Protocol

HSRP is a **First Hop Redundancy Protocol (FHRP)** that allows multiple physical routers to share a single Virtual Gateway IP. It ensures hosts have a consistent path even if a router fails.

Operational Roles

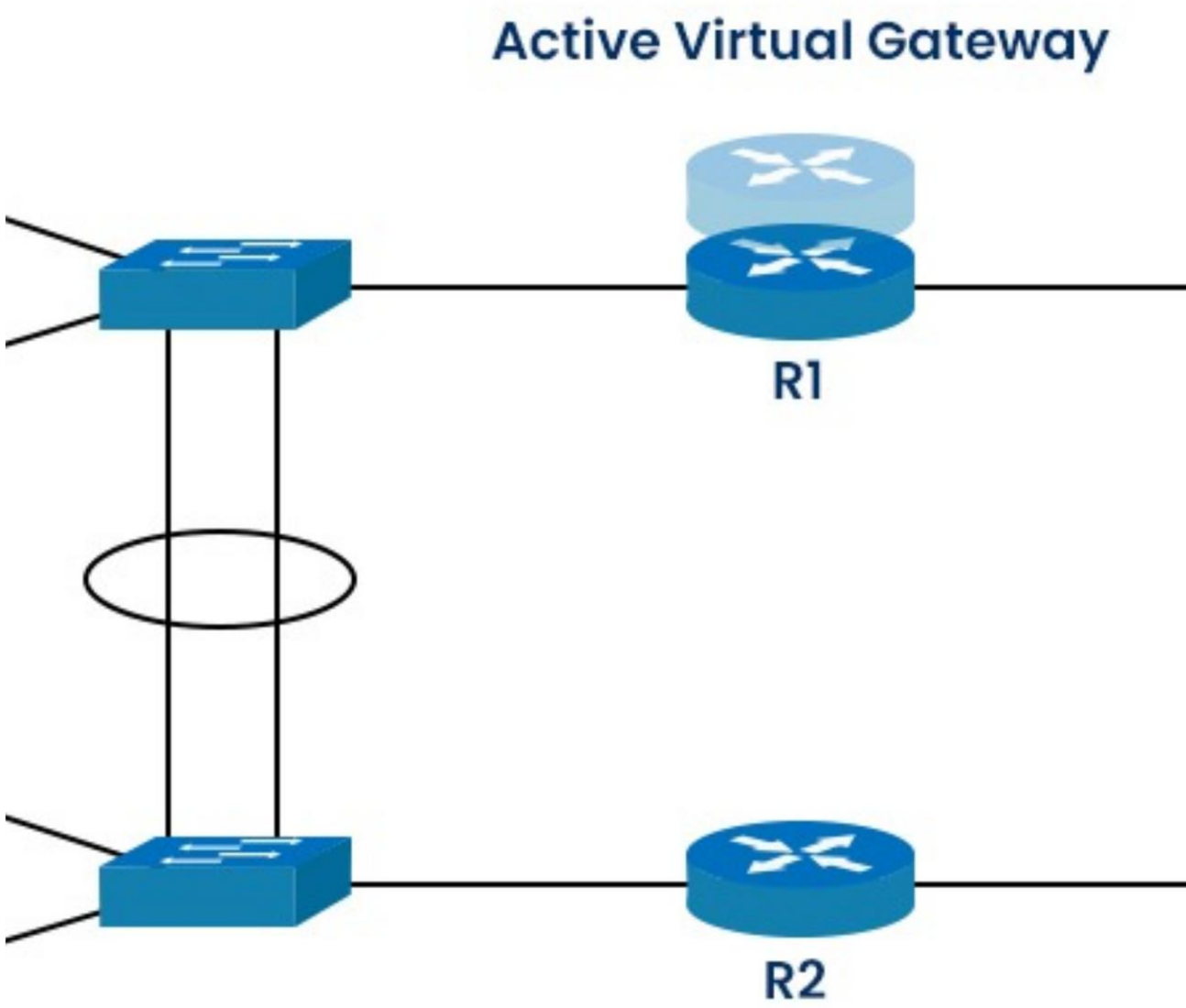
1. Active Router

The designated router that currently forwards all traffic sent to the virtual gateway address.

2. Standby Router

The "Hot" backup. It listens to **Hello** packets from the Active router and is ready to take over in milliseconds.

(Multicast Address: 224.0.0.2)



Virtual IP & MAC

Transparent Failover

Clients are configured with a single **Virtual IP (VIP)** as their gateway. HSRP handles the hardware transition behind this IP.

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Virtual MAC Address Format:
0000.0c07.acXX
```

(Where XX is the HSRP Group Number)

HSRP Election & Logic

Priority

Default is 100. The router with the **highest priority** wins the Active role election.

Preemption

Ensures a high-priority router automatically takes back the Active role after a recovery.

Timers

Hello: 3 Seconds
Hold: 10 Seconds (Failure detection interval)

Transition Timeline

Initial
Router interface starts up.

Listen
Listening for Hellos.

Speak
Participating in election.

Standby
Candidate for Active.

Active
Handling all Traffic.

FHRP Comparison Matrix

Feature	HSRP	VRRP	GLBP
Owner	Cisco Proprietary	Open Standard	Cisco Proprietary
Roles	Active / Standby	Master / Backup	AVG / AVF
Load Balancing	Manual (Multi-Group)	Manual	Automatic
Hellos	3 Seconds	1 Second	3 Seconds

Any Questions?

Thank you for your valuable time and attention.

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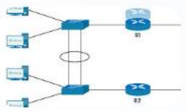

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